7. option(b)

$$\text{rate} = k[A]^n$$

$$\text{rate} = \frac{-d[A]}{dt}$$

$$\text{unit of rate} = \frac{\text{mol L}^{-1}}{s} = \text{mol L}^{-1}s^{-1}$$

$$\text{unit of rate constant} = \frac{\left(\text{mol L}^{-1}s^{-1}\right)}{\left(\text{mol L}^{-1}\right)^n} = \text{mol}^{1-n}\ \text{L}^{n-1}s^{-1}$$

in this case

$$\text{rate} = k[\text{Acetone}]^{3/2}$$

$$n = 3/2$$

$$\text{mol}^{1-(3/2)}\ \text{L}^{(3/2)-1}s^{-1}$$

$$\text{mol}^{-(1/2)}\ \text{L}^{(1/2)}s^{-1}$$

8. option(b)

A catalyst provides a new path to the reaction with low activation energy. i.e., it lowers the activation energy.

9. option(a)

In zero order reactions, increase in the concentration of reactant does not alter the rate. So statement (i) is wrong.

$$k = A\, e^{-\left(\frac{E_a}{RT}\right)}$$

if $E_a = 0$ so, statement (ii) is correct, and statement (iii) is wrong

$$k = A\, e^0$$

$$k = A$$

$$\ln k = \ln A - \left(\frac{E_a}{R}\right)\left(\frac{1}{T}\right)$$

this equation is in the form of a straight line equatoion

$$y = c + m\, x$$

a plot of $\ln k$ vs $\left(\frac{1}{T}\right)$ is a straight line with negative slope

so statements (iv) and (v) are wrong.

Answers.indd 245 3/29/2019 6:44:00 PM